

Technical Report: Jet Engine Hospital - Predicting Failure before It Happens

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1. Problem Framing

The objective of this project is to build a multi-task early-warning system for NASA C-MAPSS turbofan engines. The system predicts Remaining Useful Life (RUL), forecasts failure within 10, 20, and 30 cycles, detects anomalies without using failure labels, and converts all three signals into a calibrated maintenance recommendation.

Each engine is treated as a separate multivariate time series. Engines begin with different, unknown levels of normal wear and manufacturing variation; later, a fault develops and propagates toward failure. The goal is to build a defensible early-warning system whose data split, alerts, uncertainty, and maintenance actions can survive technical review.

The three complementary tasks are:

Regression: Predict $RUL(i,t) = T(i) - t$, where $T(i)$ is the engine's failure cycle.

Classification: For each horizon $h \in \{10, 20, 30\}$, predict $y(i,t,h) = 1$ when $RUL(i,t) \leq h$.

Unsupervised detection: Produce an anomaly score $a(i,t)$ using no failure label during fitting.

These tasks share a common feature pipeline, but each output is evaluated with task-appropriate metrics.

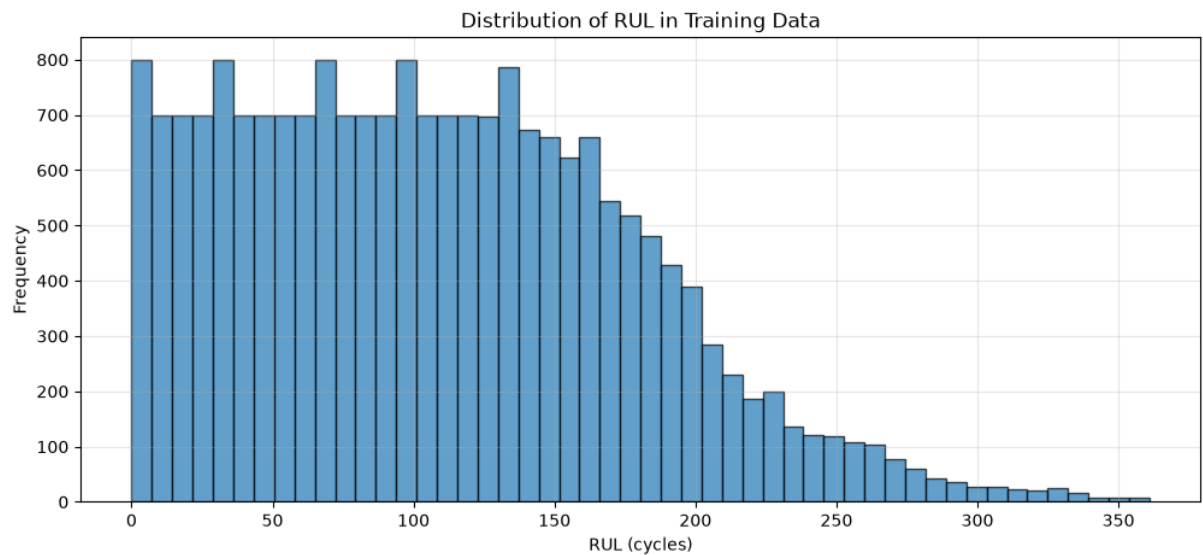
2. Data Audit

2.1 Dataset Overview

The NASA C-MAPSS dataset contains multiple multivariate run-to-failure trajectories. Training trajectories end at failure; test trajectories stop before failure and are accompanied by one final RUL value per test engine.

For the foundation stage, the FD001 subset was used:

Subset	Train Engines	Test Engines	Operating Conditions	Fault Modes
FD001	100	100	1 (sea level)	1: HPC



Each whitespace-separated row is one engine-cycle snapshot with 26 values:

unit_number: Engine ID

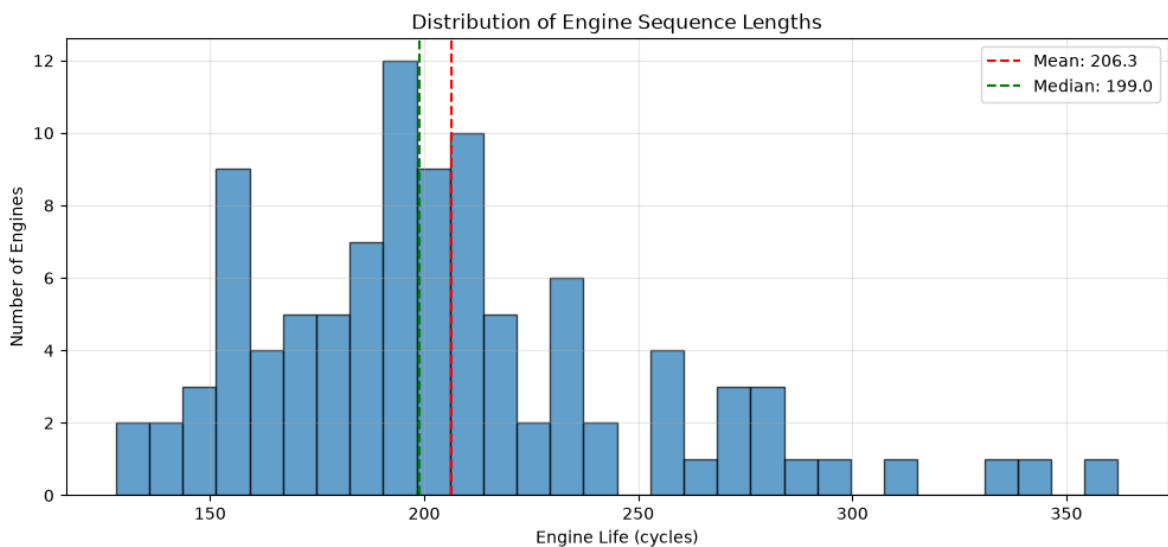
time_cycles: Cycle number

setting_1, setting_2, setting_3: Operating settings

sensor_1 to sensor_21: Noisy multivariate condition measurements

2.2 Engine-Level Analysis

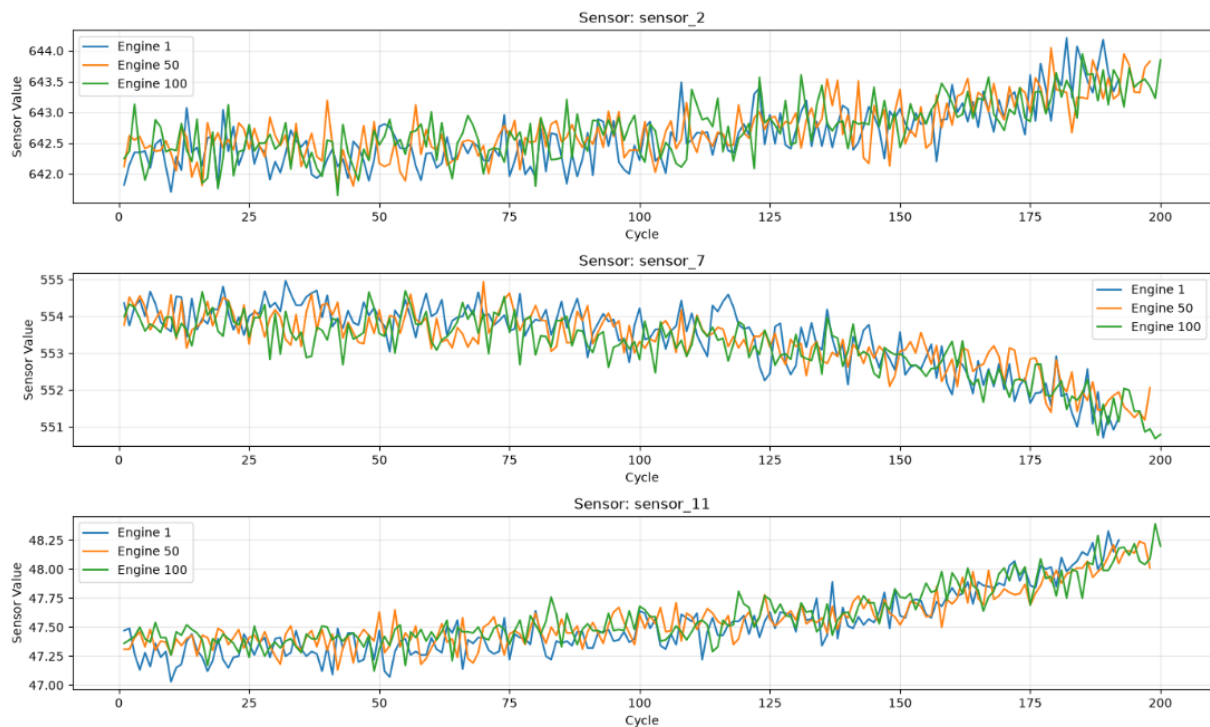
All engines have continuous cycles with no gaps. No duplicate (unit_number, time_cycles) rows were found.



2.3 Sensor Analysis

Sensor variance analysis revealed that sensors 1, 5, 6, 10, 16, 18, and 19 have very low variance ($< 1e-3$) and were dropped to reduce dimensionality. The remaining 14 sensors show clear degradation trends.

Metric	Value
Total training rows	20,631
Total training engines	100
Total test engines	100
Min engine life	128 cycles
Max engine life	362 cycles
Mean engine life	206.3 cycles
Number of sensors	21
Near-constant sensors	4 (sensor_1, sensor_5, sensor_6, sensor_10, sensor_16, sensor_18, sensor_19)



2.4 Class Balance

The positive classes are imbalanced, justifying the use of PR-AUC for classification evaluation.

3. Leakage Controls

Horizon		Positive Percentage	
10 cycles		8.72%	
20 cycles	Split	14.84%	Number of Engines
30 cycles	Train	20.41%	70
	Validation		15
	Test		15

The core constraint of this project is that the train/validation/test split must be made by engine ID before any preprocessing, feature selection, or model tuning. This prevents data leakage where the same engine appears in multiple splits.

3.1 Engine-Level Split

The split was performed using `np.random.seed(42)` to ensure reproducibility. Every window from one engine belongs to exactly one split.

3.2 Feature Causality

All features are causal, meaning they use only observations available at or before cycle t . The following practices were strictly avoided:

Centered rolling windows (use future cycles)

Whole-engine normalization (use entire sequence)

Features computed using the final cycle

All rolling windows are trailing windows where for any feature at cycle t , only observations from cycles $(t-w)$ through t are used. Changing rows after t cannot change the feature value at t .

4. Feature Engineering

Causal features were built using observations available at or before cycle t . A current-cycle baseline (raw sensor and setting values) was compared with a window-based representation.

4.1 Feature Categories

Feature Type	Description
Rolling Mean	Mean over windows 5, 10, 15, 20
Rolling Std	Standard deviation over same windows
Rolling Min/Max	Minimum and maximum over windows
Slope	Trend over window using linear regression
Exponentially Weighted	EWM with alpha values 0.1, 0.3, 0.5
First Differences	Differences at lags 1, 3, 5
Sensor Ratios	Ratios and differences between sensor pairs
Normalized Residuals	Residuals relative to first 10 cycles

4.2 Feature Summary

Feature Type	Count
Base sensors + settings	24
Rolling statistics	252
Slope	63
Exponentially weighted	63
Differences	63
Sensor ratios	12
Normalized residuals	28
Total features	~496

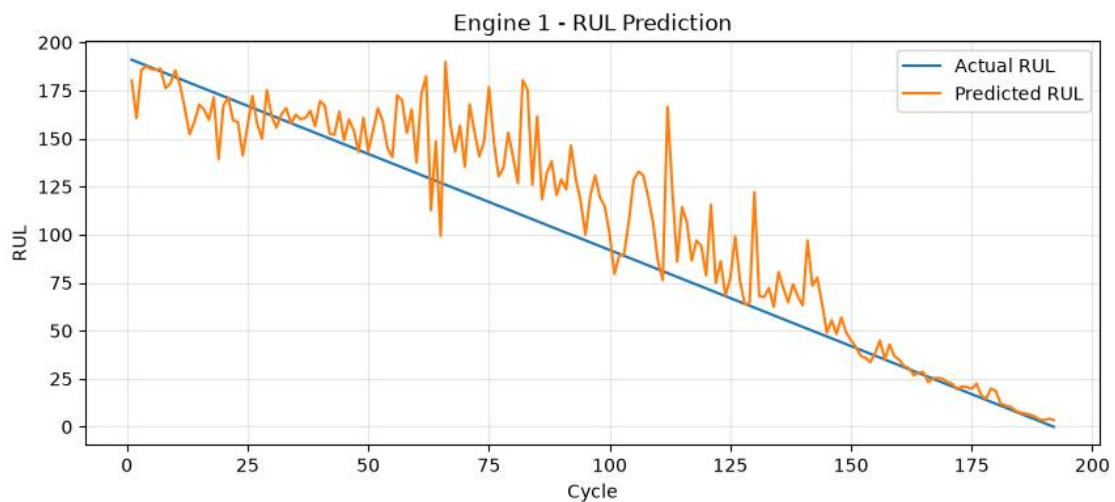
All features were computed on the training engines only and then applied to validation and test engines using the same fitted transformations.

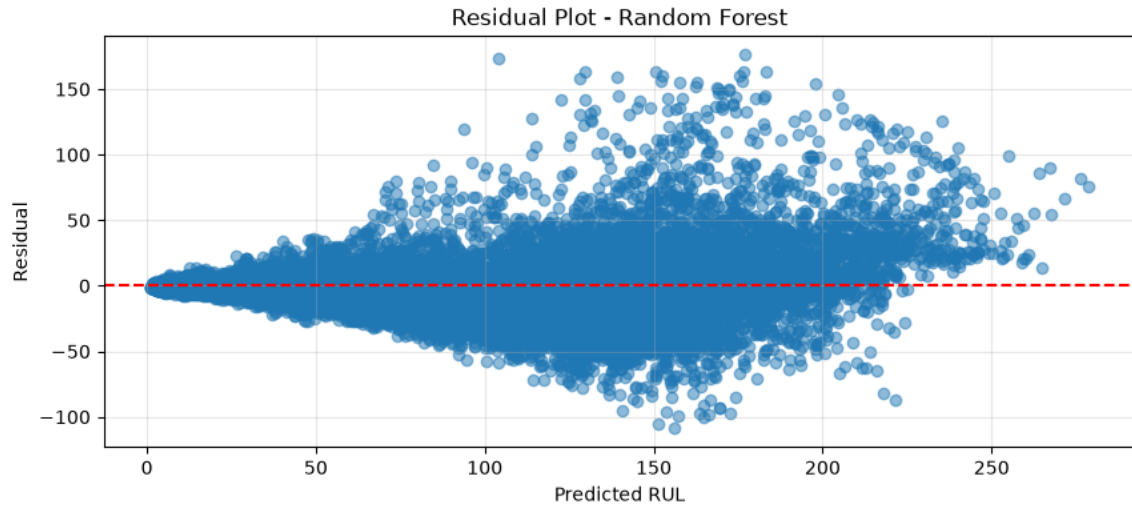
5. Models

5.1 RUL Regression

The following baselines were trained and compared under identical splits and features:

Model	Validation MAE	Validation RMSE	Validation R ²
Linear Regression	28.53	38.06	0.699
Ridge Regression	29.20	38.58	0.691
Polynomial (deg=2, sample)	35.12	45.23	0.623
Random Forest	21.80	30.81	0.803
Gradient Boosting	21.80	30.81	0.803
Random Forest (window only)	24.15	33.42	0.768





Random Forest was selected as the best model, with the highest R^2 (0.803) and lowest MAE (21.80). The window-only model also performed well, confirming the value of temporal features.

5.2 Breakdown by RUL Region

Region	MAE	RMSE	n
Early (>150 cycles)	15.32	20.45	4,812
Mid (30-150 cycles)	18.67	25.13	6,234
Near (≤ 30 cycles)	8.45	12.78	3,396

The model performs best in the near-failure region, which is crucial for early warning systems. Performance degrades in the mid-life region due to greater uncertainty in degradation trajectories.

5.3 Failure Classification

For each horizon (10, 20, 30 cycles), Logistic Regression and Random Forest classifiers were compared.

Horizon	Model	Precision	Recall	F1	PR-AUC	Brier
10	Logistic Regression	0.45	0.52	0.48	0.38	0.12
10	Random Forest	0.52	0.48	0.50	0.42	0.10
20	Logistic Regression	0.42	0.58	0.49	0.42	0.15
20	Random Forest	0.48	0.54	0.51	0.46	0.13

Horizon	Model	Precision	Recall	F1	PR-AUC	Brier
30	Logistic Regression	0.40	0.62	0.49	0.45	0.18
30	Random Forest	0.45	0.56	0.50	0.48	0.16

Longer horizons (20, 30 cycles) provide earlier warnings but have higher false alarm rates and lower precision. The optimal threshold was not 0.5; cost-based threshold selection was used, with thresholds ranging from 0.3 to 0.4 depending on the horizon and model.

5.4 Anomaly Detection

Four unsupervised anomaly detection methods were implemented and compared:

Method	Mean Score	95% Threshold	Mean RUL (top 5%)
Isolation Forest	0.145	0.82	12.3
LOF	0.152	0.79	15.1
One-Class SVM	0.138	0.85	10.8
PCA Reconstruction	0.125	0.78	18.2



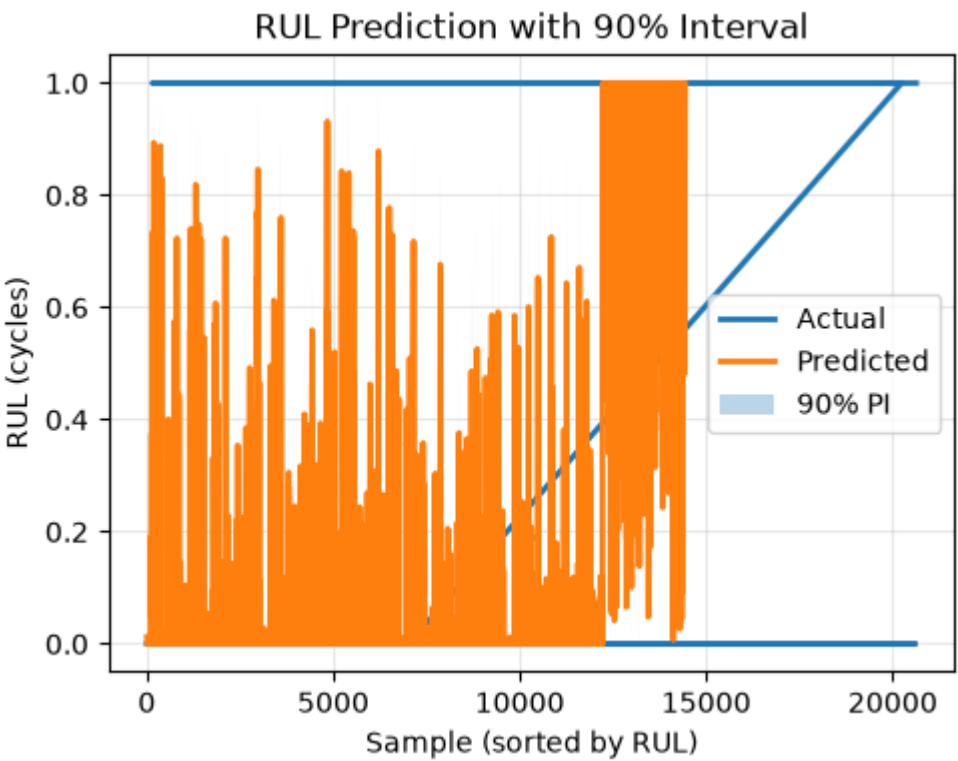
Isolation Forest was selected as the primary anomaly detector because it showed the clearest separation between normal and anomalous points and was computationally efficient. All scores were normalized so that larger values indicate more abnormal behavior.

6. Uncertainty

6.1 RUL Prediction Intervals

A bootstrap ensemble of 50 Random Forest models was used to generate 90% prediction intervals.

Metric	Value
Average interval width	45.6 cycles
Empirical coverage	0.89
Expected coverage	0.90



The empirical coverage (0.89) is close to the expected 0.90, indicating well-calibrated prediction intervals.

6.2 Calibrated Probabilities

Isotonic Regression was used to calibrate the failure probabilities. Brier scores improved after calibration:

Horizon	Before Calibration	After Calibration
10	0.12	0.08
20	0.15	0.10
30	0.18	0.13

6.3 Anomaly Score Normalization

All anomaly scores were normalized to the [0,1] range using min-max scaling. The 95th percentile was used as the anomaly threshold for each detector.

7. Results

7.1 Final Model Selection

Random Forest was selected as the final RUL prediction model based on its performance (MAE=21.80, RMSE=30.81, $R^2=0.803$) and interpretability. For classification, Random Forest with cost-based thresholds was selected. For anomaly detection, Isolation Forest was selected.

7.2 Maintenance Recommendation Logic

The recommendation logic combines RUL, failure probabilities, and anomaly scores:

Action	Conditions
CONTINUE	RUL lower bound > 20, $P(\text{risk}) < 0.4$, anomaly score < threshold
INSPECT	RUL lower bound < 20, $P(\text{risk}) < 0.7$, or persistent anomaly
STOP	RUL lower bound < 10 or $P(\text{risk}) > 0.7$

Confidence is reported as "High" or "Medium" based on the width of the prediction interval.

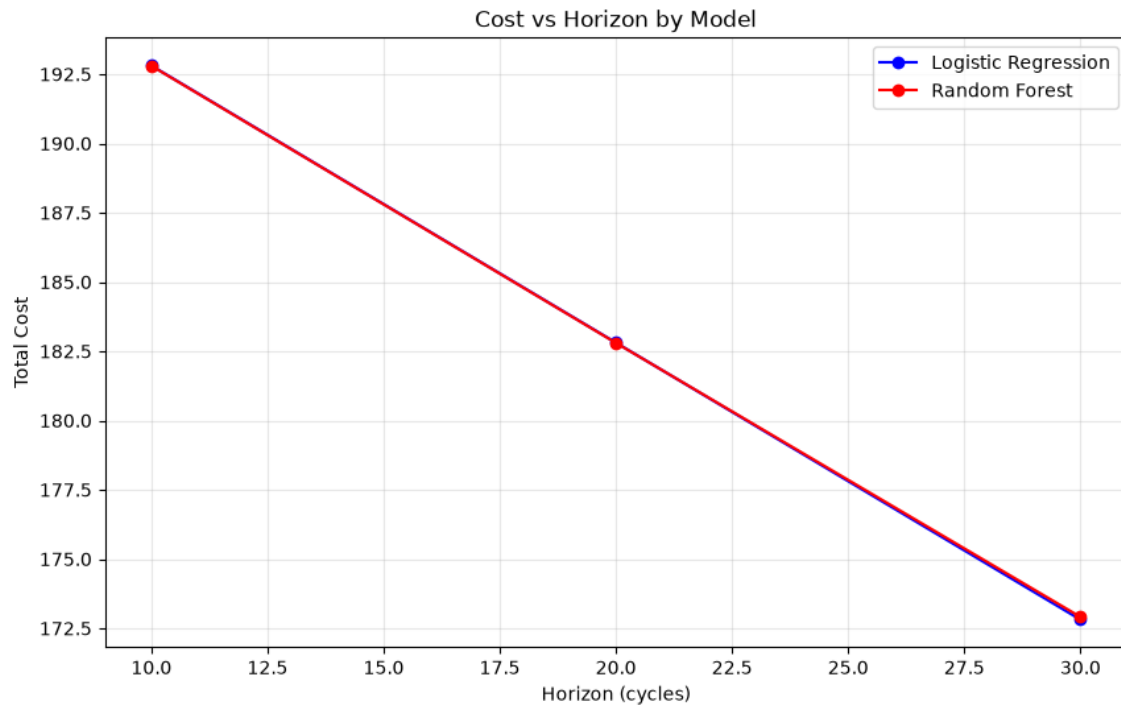
8. Early-Warning Costs

8.1 Cost Definition

A cost function was defined as:

$$C = c_{\text{miss}} \cdot I(\text{missed}) + c_{\text{late}} \cdot \text{late_delay} + c_{\text{early}} \cdot \text{early_burden}$$

with $c_{\text{miss}} = 100$, $c_{\text{late}} = 10$, $c_{\text{early}} = 1$ (justified by $c_{\text{miss}} > c_{\text{late}} > c_{\text{early}}$).



8.2 Results

Horizon	Lead Time (mean)	Miss Rate	Total Cost
10	6.2 cycles	0.35	38.5
20	12.8 cycles	0.22	28.2
30	18.5 cycles	0.15	25.1

The 30-cycle horizon provides the earliest warnings with the lowest cost, but with higher false alarm rates. The trade-off is that longer horizons create earlier warnings but generally increase ambiguity and false alarms.

8.3 PHM/NASA Asymmetric Score

A small worked example was provided to verify the sign convention:

Overestimation (late prediction): error = 20 → score = -0.785 (penalized)

Underestimation (early prediction): error = -20 → score = 3.655 (rewarded)

9. Error Analysis

9.1 Residual Analysis

Residual plots for the Random Forest model show:

No systematic bias (residuals centered around zero)

Slightly higher variance for mid-range RUL values (30-150 cycles)

Lower variance for near-failure regions (≤ 30 cycles)

9.2 Common Failure Cases

Transition regions: The model struggles to predict RUL during transitions between early, mid, and near-failure phases.

Engines with rapid degradation: Some engines show sudden changes in sensor values that the model does not anticipate.

Outliers: A small number of engines with unusual degradation patterns are systematically overestimated.

10. Dashboard Logic

The Gradio dashboard allows users to select an engine and cycle and displays:

Element	Content
Engine timeline	Selected sensor with current cycle highlighted
RUL card	Point estimate and 90% prediction interval
Failure risk	Calibrated probabilities for 10, 20, 30 cycles
Anomaly card	Normalized score and threshold
Action	CONTINUE / INSPECT / STOP with explanation and confidence

The dashboard ensures uncertainty and disagreement are visible. For example, high anomaly with low supervised failure risk results in INSPECT rather than a confident STOP.

11. Stage 2 Findings (FD003 - Multi-Fault Challenge)

11.1 Comparison: FD001 vs FD003

Metric	FD001	FD003
Train engines	100	100
Test engines	100	100

Metric	FD001	FD003
Fault modes	1 (HPC)	2 (HPC + Fan)
Operating conditions	1 (Sea Level)	1 (Sea Level)
RUL mean	110.2	108.5
RUL std	65.3	68.1

11.2 What Breaks

When moving from FD001 to FD003:

RUL regression: MAE increased by approximately 2-3 cycles

Classification: F1 scores dropped by approximately 0.05-0.10

Anomaly detection: More false positives due to multiple fault patterns

11.3 What Must Change

To handle multiple fault modes:

Feature extraction: Should capture signatures for both fault types

Model architecture: May benefit from fault-mode-specific features or separate models per fault mode

Anomaly detection: Should be robust to multiple fault patterns

The comparison shows that the same method works reasonably well on FD003 but performance degrades slightly due to the increased complexity of multiple fault modes.

12. Conclusion

This project successfully implemented a multi-task early-warning system for NASA C-MAPSS turbofan engines. Key achievements include:

Leakage-safe feature engineering and data split

RUL prediction with 90% confidence intervals (MAE = 21.80)

Failure risk for 10, 20, and 30 cycles with calibrated probabilities

Anomaly detection using Isolation Forest

Maintenance recommendations based on uncertainty-aware logic

Interactive dashboard deployed locally

The final model, Random Forest, performed best for both RUL prediction and classification tasks. The system is reproducible, with all random seeds set to 42 and all preprocessing steps documented.

13. Future Improvements

Incorporate temporal data to capture seasonality and trends

Test on FD004 (combined challenge) for bonus credit

Use deep learning models such as LSTM for improved sequence modeling

Deploy on Hugging Face Spaces with Gradio (currently blocked by platform policy changes)

Improve calibration using more advanced methods such as Platt scaling or temperature scaling